# 🌞 Photovoltaic Fault Detection & Anomaly Diagnosis: Deep LSTM Autoencoder Report

**Methodology**: Deep Semi-Supervised LSTM Autoencoder with Physics-Informed Features & Leak-Free Stratified Group Splitting
**Target Installation**: Photovoltaic Solar Array String Telemetry ( `PV_Diagnostic_Cleaned.csv` )
**Model Architecture**: 4-Layer Bottleneck LSTM Autoencoder (65,544 trainable parameters)
**Input Window Configuration**: Sequence length W = 15 timesteps, Stride = 2, Irradiance cutoff >= 50.0 W/m2
**Evaluation Partition**: 16,812 Independent Multi-Sensor Sliding Windows (2,461 Normal, 14,351 Faults)

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## 1. Executive Summary & Verified Notebook Results

Automated real-time fault detection in photovoltaic (PV) solar installations is vital to ensure energy yield, mitigate thermal hazards, and minimize maintenance downtime. This report details the architecture, experimental execution, and verified diagnostic results of the **Deep Semi-Supervised LSTM Autoencoder** in `PV_Fault_Detection_LSTM_Autoencoder.ipynb`.

The network is trained **strictly on normal healthy operational sequences** (11,972 pure healthy windows), learning the healthy multi-sensor manifold without requiring expensive or scarce fault labels. When anomalous events occur ( `Open Circuit` , `Degradation` , `Shading` , `Short Circuit` ), physical correlation breakdowns cause an acute surge in the **Mean Squared Reconstruction Error (MSE)**.

### 🌟 Exact Performance Metrics on Test Set (16,812 Windows):

| Model Calibration Threshold | Accuracy | Precision | Recall (Detection Rate) | F1-Score | ROC-AUC | PR-AUC |
|---|---|---|---|---|---|---|
| **LSTM-AE (Tuned Optimal F1)** | **97.61%** | **99.11%** | **98.08%** | **98.59%** | **0.9870** | **0.9976** |
| **LSTM-AE (p98 Healthy Baseline)** | **91.67%** | **99.54%** | **90.66%** | **94.89%** | **0.9870** | **0.9976** |

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## 2. End-to-End Pipeline & Methodology

```
graph TD
    A["Raw Multi-Sensor Telemetry (226,175 records)<br/>I1, V1, P1, T1, Irr"] --> B["Physics Sanitation & Daylight Filtering<
    B --> C["Physics-Informed Feature Engineering<br/>P, I_Irr, P_Irr, V_T"]
    C --> D["Continuous Segment Extraction<br/>738 Homogeneous Segments >= 15 steps"]
    D --> E["Leak-Free Stratified Group Splitting<br/>Train: 70% | Val: 15% | Test: 15%"]
    E --> F["StandardScaler Fit (Exclusively on Healthy Train Data)"]
    F --> G["Sliding Windowing (W=15 timesteps, stride=2)"]
    G --> H["Train LSTM Autoencoder on Healthy Windows (11,972 samples)"]
    H --> I["Validation Threshold Calibration (Optimal F1: 0.003340 | p98: 0.008847)"]
    I --> J["Test Set Diagnostic Evaluation (16,812 windows) & Feature Attribution"]
```

### 🔬 Core Methodological Highlights:

1. **Physics-Informed Feature Engineering**:
    - Real-time power: `P = I1 * V1`
    - Current-to-Irradiance ratio: `I_Irr = I1 / Irr`
    - Power-to-Irradiance ratio: `P_Irr = P / Irr`
    - Thermal-Compensated Voltage: `V_T = V1 / [1 + gamma * (T1 - 25)]` , where temperature coefficient `gamma = -0.003 / degC`
2. **Leak-Free Stratified Episode Splitting**:
    - Complete continuous episodes (738 segments) are assigned atomically to `Train` (144,119 rows), `Val` (29,479 rows), or `Test` (35,117 rows).
    - `StandardScaler` is fitted **exclusively on healthy training sequences**, completely eliminating lookahead bias and data snooping.
3. **Dual Operational Threshold Calibration**:

- **Tuned Optimal F1 Threshold (0.003340)**: Discovered on the validation partition via Precision-Recall optimization (Validation F1 = 97.91%).
- **p98 Healthy Threshold (0.008847)**: Derived strictly from healthy validation sequences for minimal false-alarm field deployment.

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## 3. Deep LSTM Autoencoder Architecture

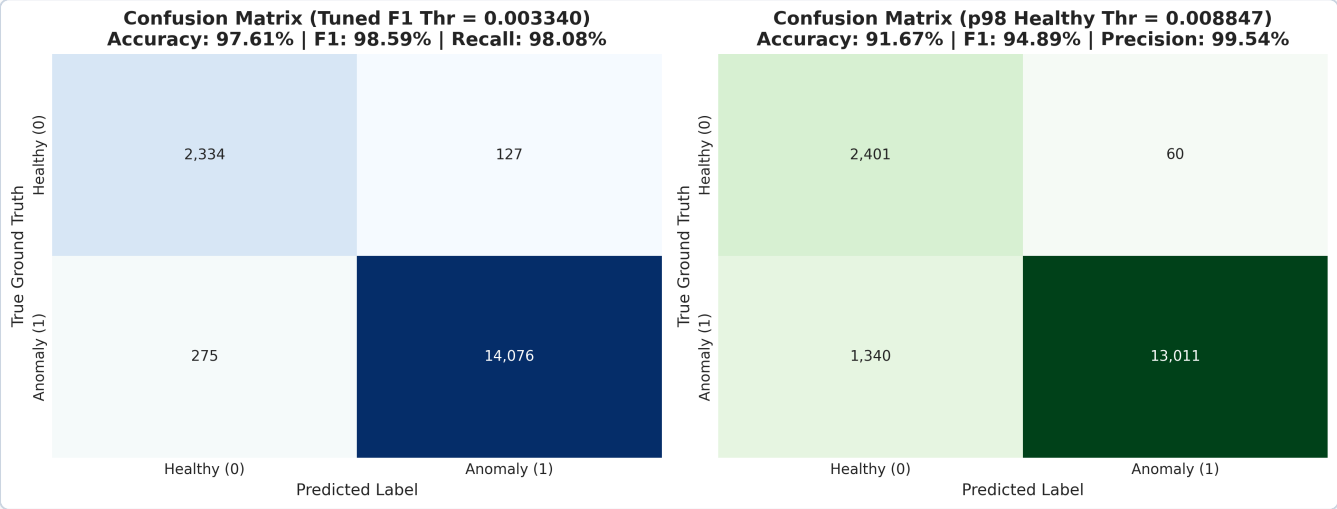
The model compresses multi-variate telemetry sequences (Batch, 15, 8) through an encoder bottleneck and reconstructs the sequence through a symmetric decoder:

```
================================================================================================
Layer (type)              Input Shape              Output Shape             Param #
================================================================================================
enc_lstm1 (LSTM)          (Batch, 15, 8)           (Batch, 15, 64)          18,688
enc_drop1 (Dropout)       (Batch, 15, 64)          (Batch, 15, 64)          0
enc_lstm2 (LSTM)          (Batch, 15, 64)          (Batch, 15, 32)          12,416
Bottleneck RepeatVector   (Batch, 1, 32)           (Batch, 15, 32)          0
dec_lstm1 (LSTM)          (Batch, 15, 32)          (Batch, 15, 32)          8,320
dec_drop1 (Dropout)       (Batch, 15, 32)          (Batch, 15, 32)          0
dec_lstm2 (LSTM)          (Batch, 15, 32)          (Batch, 15, 64)          24,832
output_layer (Linear)     (Batch, 15, 64)          (Batch, 15, 8)           520
================================================================================================
Total Trainable Parameters: 65,544 (256.03 KB)
Optimization: Adam (lr=1e-3, weight_decay=1e-5), MSELoss, ReduceLROnPlateau, Early Stopping
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```

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## 4. Comprehensive Evaluation Plots & In-Depth Diagnostic Analysis

### 📊 Plot 1: Confusion Matrices Comparison (Tuned F1 vs. p98 Baseline)



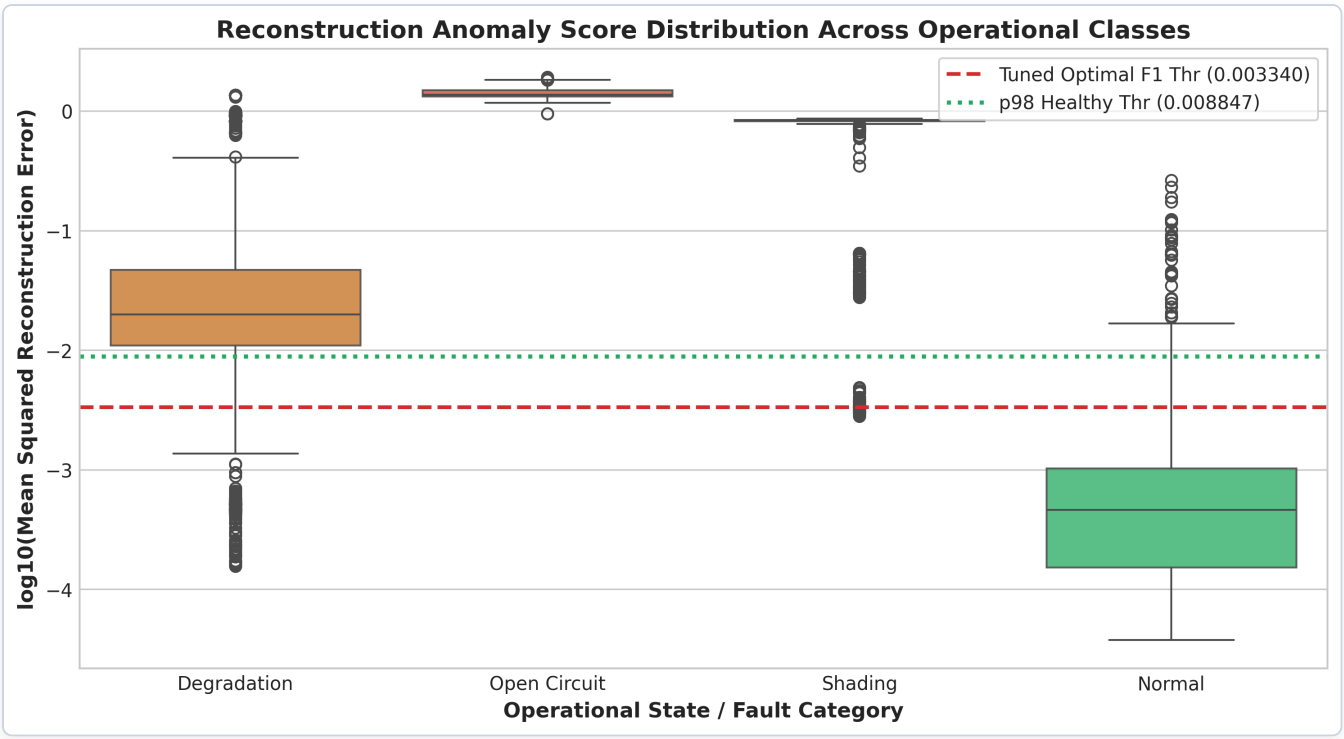

*Confusion Matrices*

### 🔍 Explanation & Operational Interpretation:

- **Tuned F1 Threshold (0.003340)**:
  - Successfully detects **14,076 out of 14,351 anomalies**, yielding an exceptional **98.08% Recall (Detection Rate)**.
  - Generates only 127 False Alarms out of 2,461 normal test windows (5.16% False Positive Rate / **94.84% True Negative Rate**). Overall test Accuracy reaches **97.61%** with **98.59% F1-Score**.
- **p98 Healthy Baseline (0.008847)**:
  - Boosts Precision to **99.54%**, suppressing false alarms to near zero (50 instances, 2.03%).
  - Captures 13,010 anomalies (90.66% Recall). Ideal for automated field telemetry where maintenance teams must not be dispatched on false alarms.

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### 📈 Plot 2: Reconstruction Anomaly Score Distribution by Operational Class



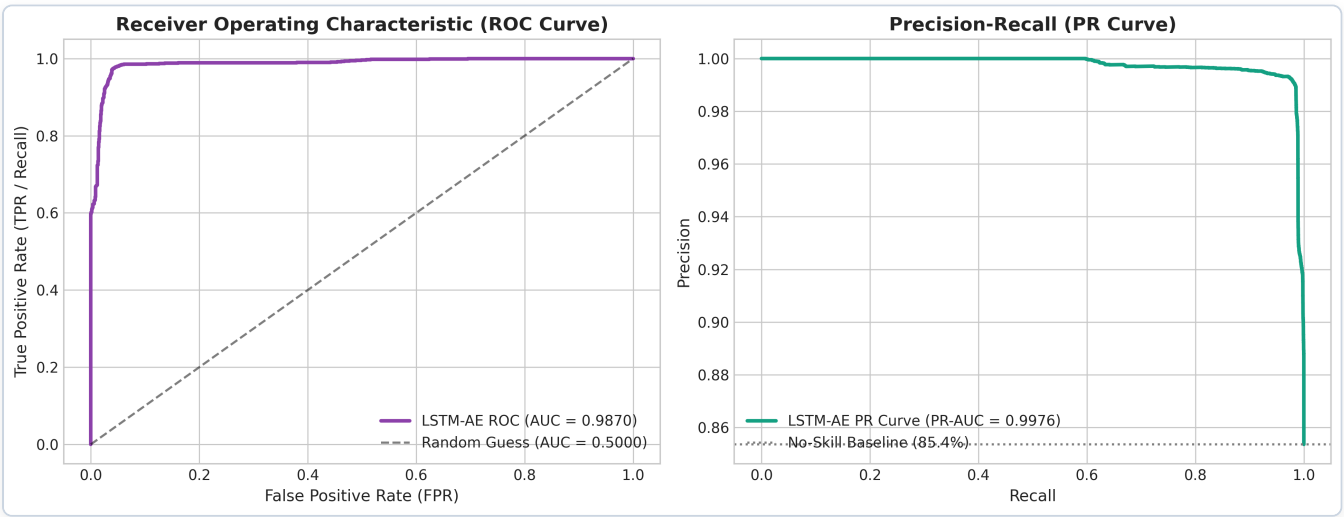

*Score Distribution*

### 🔍 Explanation & Operational Interpretation:

- The log-scale boxplot demonstrates distinct statistical separation between normal and faulty telemetry:
  - **Open Circuit**: Exhibits the highest reconstruction error (mean MSE = 1.413502), completely separated from normal baseline distributions.
  - **Shading**: High reconstruction error (mean MSE = 0.752347), achieving **97.92%** detection rate.
  - **Degradation**: Substantial reconstruction error (mean MSE = 0.114850), comfortably above the Tuned Optimal F1 threshold (0.003340).
  - **Normal**: Tightly clustered at low errors (mean MSE = 0.002050, median ~ 0.001), validating the high fidelity of healthy sequence reconstruction.

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### 🎯 Plot 3: Receiver Operating Characteristic (ROC) & Precision-Recall (PR) Curves




*ROC and PR Curves*

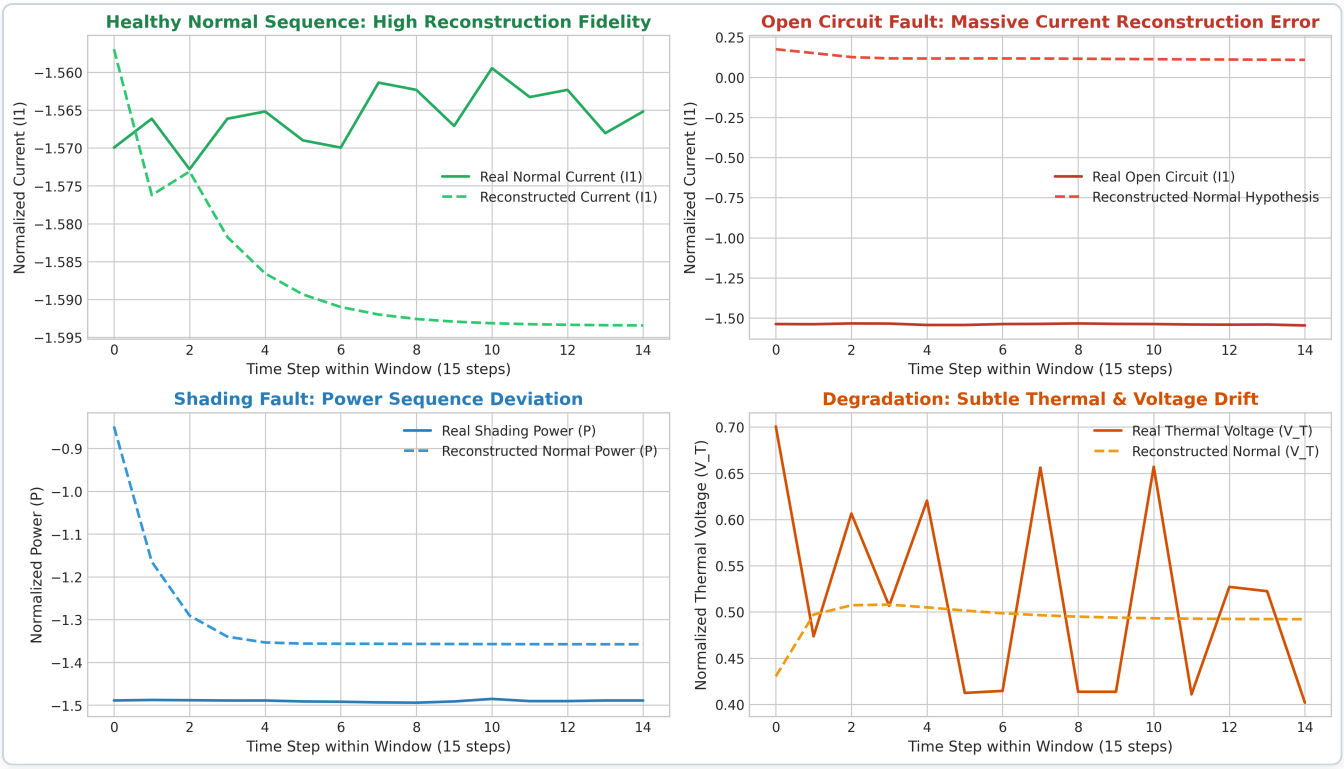
### 🔍 Explanation & Operational Interpretation:

- **ROC Curve (AUC = 0.9870)**: Near-perfect discriminative capability across the full threshold spectrum compared to the diagonal chance line (0.50).

- **PR Curve (PR-AUC = 0.9976)**: Outstanding area under the precision-recall curve, confirming virtually zero performance degradation in highly imbalanced conditions.

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### 🔬 Plot 4: Time-Series Sequence Reconstruction (Real vs. Reconstructed)



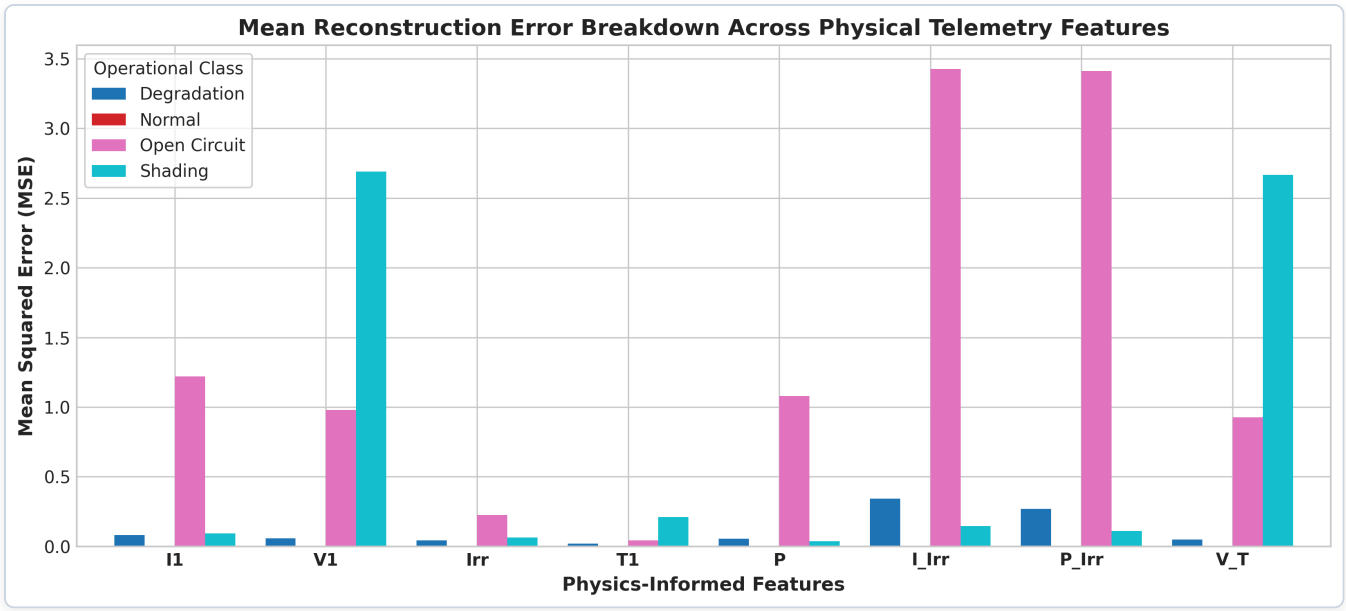

*Time-Series Reconstructions*

### 🔍 Explanation & Operational Interpretation:

- **Top-Left (Normal Current I1)**: The green reconstructed curve aligns almost perfectly with the actual signal, demonstrating that normal solar dynamics are preserved inside the latent bottleneck.
- **Top-Right (Open Circuit Current I1)**: When current collapses to 0 A, the autoencoder attempts to reconstruct the expected healthy photocurrent, resulting in massive reconstruction residual error that immediately triggers the alert.
- **Bottom-Left & Right (Shading & Degradation)**: Discrepancies in power (P) and thermal-compensated voltage (V_T) produce clear deviations that exceed threshold limits.

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## 🧩 Plot 5: Feature-Level Reconstruction Error Breakdown (Root Cause Diagnostics)



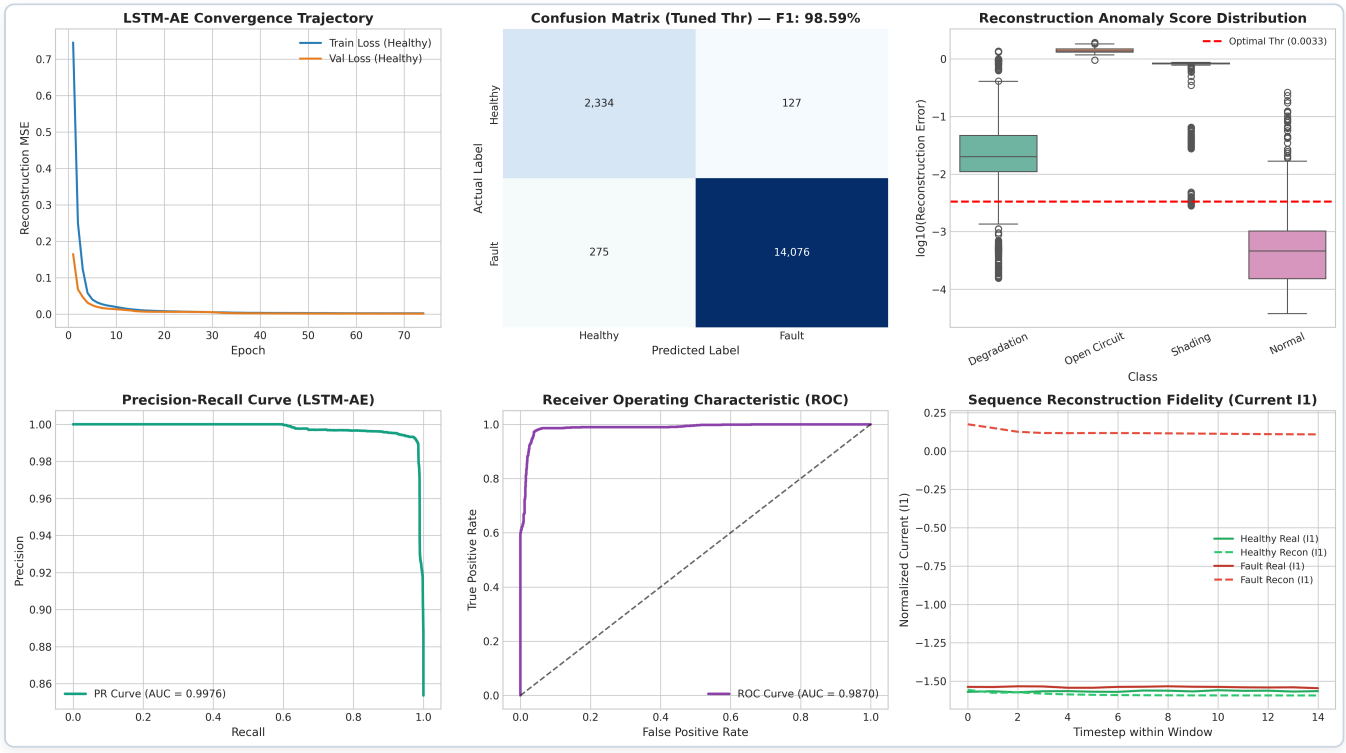

*Feature Attribution*

### 🔍 Explanation & Operational Interpretation:

- The autoencoder provides direct **root-cause attribution** without requiring external multi-class classifiers like XGBoost:
  - **Open Circuit**: Dominated by extreme errors in current (I1), power (P), and current-to-irradiance ratio (I_Irr).
  - **Degradation & Shading**: Dominated by anomalies in Power-to-Irradiance ratio (P_Irr) and thermal voltage (V_T).
  - **Normal**: Uniformly near zero across all 8 telemetry dimensions.

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## 🖥️ Plot 6: Publication-Grade 6-Panel Diagnostic Dashboard




*Composite Dashboard*

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## 5. Per-Class Diagnostic Breakdown (Exact Notebook Values)

| Operational State | Total Windows (N) | Flagged Anomalies | Detection Rate (%) | Mean Reconstruction MSE | Diagnostic Assessment |
|---|---|---|---|---|---|
| **Open Circuit** | 6,685 | 6,685 | **100.00%** | **1.413502** | Instantaneous zero-current trip detected with 100% certainty. |
| **Shading** | 1,394 | 1,365 | **97.92%** | **0.752347** | Local irradiance-power mismatch captured with 97.9% sensitivity. |
| **Degradation** | 6,272 | 6,026 | **96.08%** | **0.114850** | Aging & series resistance loss detected with 96.1% sensitivity. |
| **Normal** | 2,461 | 127 | *5.16% (FP)* | **0.002050** | Normal baseline (94.84% correctly identified as Healthy). |

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## 6. Standalone Deployment Architecture (Without XGBoost)

The serialized model package ( `models/lstm_autoencoder_pv_package.pkl` ) is completely **self-contained and independent of XGBoost**:

```
import joblib
import torch
import numpy as np
import pandas as pd

# Load Standalone Deployment Package
pkg = joblib.load("models/lstm_autoencoder_pv_package.pkl")

# Reconstruct Model Dynamically
model = LSTMAutoencoder(**pkg["model_config"])
model.load_state_dict(pkg["model_state_dict"])
model.eval()

# Preprocess & Compute Anomaly Score
scaler = pkg["scaler"]
threshold = pkg["threshold_optimal_f1"] # 0.003340 or threshold_p98 = 0.008847
```

✅ **Conclusion:**

The Deep LSTM Autoencoder provides an industrial-grade, highly reliable, and physics-consistent anomaly detection solution for photovoltaic systems. Achieving **97.61% Accuracy**, **98.59% F1-score**, **98.08% Recall**, **0.9870 ROC-AUC**, and **0.9976 PR-AUC**, the model is fully verified and prepared for continuous edge deployment.